

Soil Classification Using Atterberg Limits

Classification Logic — A Programmer's Guide

Cransfield Materials Testing Centre

Think of It as a Decision Tree

The entire classification process is essentially a series of if-else checks on two input values — LL and PL. Everything else is derived from those two numbers.

The Inputs

The two values received from the user are:

```
LL → Liquid Limit (a percentage, e.g. 48)
PL → Plastic Limit (a percentage, e.g. 22)
```

The first derived value is the Plasticity Index:

```
PI = LL - PL → (e.g. 48 - 22 = 26)
```

PI is not a separate input — it is always calculated. Never accept it directly from the user without verifying it matches LL minus PL.

Step Three — The A-Line Decision

The A-line is simply a formula that returns a threshold value of PI for any given LL:

```
A_line_value = 0.73 × (LL - 20)
```

You then compare the sample's actual PI against this threshold:

```
if PI > A_line_value → soil is CLAY → first letter = "C"
if PI < A_line_value → soil is SILT → first letter = "M"
if PI = A_line_value → treat as CLAY by convention
```

This is the first branch in the decision tree. It gives you either C or M — nothing else.

Step Four — The Plasticity Descriptor

Once you have C or M, you look up the second letter by checking which range the LL falls into. Think of it like a lookup table:

```
if LL < 35          → second letter = "L"  (Low)
if 35 ≤ LL < 50     → second letter = "I"  (Intermediate)
if 50 ≤ LL < 70     → second letter = "H"  (High)
if 70 ≤ LL < 90     → second letter = "V"  (Very High)
if LL ≥ 90          → second letter = "E"  (Extremely High)
```

This is the second branch. No formula needed — just range checking.

Step Five — The Final Classification

Once you have both letters, the classification is simply a string concatenation:

```
BS_classification = first_letter + second_letter
```

Examples:

Input	Result	Description
C + I	CI	Clay, Intermediate plasticity
C + H	CH	Clay, High plasticity
M + L	ML	Silt, Low plasticity
C + V	CV	Clay, Very High plasticity
M + H	MH	Silt, High plasticity

USCS Classification — Even Simpler

USCS uses only one boundary for LL (at 50%) and the same A-line check. There are only four possible outputs:

```
if LL < 50 and PI ≥ A_line_value → "CL"
if LL ≥ 50 and PI ≥ A_line_value → "CH"
if LL < 50 and PI < A_line_value → "ML"
if LL ≥ 50 and PI < A_line_value → "MH"
```

Four possible outputs. Four conditions. Done.

The Complete Logic Flow

```
START
|
|— Calculate  $PI = LL - PL$ 
|
|— Calculate  $A\_line\_value = 0.73 \times (LL - 20)$ 
|
|— CHECK:  $PI$  vs  $A\_line\_value$ 
|   |—  $PI \geq A\_line\_value \rightarrow first\_letter = "C"$ 
|   |—  $PI < A\_line\_value \rightarrow first\_letter = "M"$ 
|
|— CHECK:  $LL$  range
|   |—  $LL < 35 \rightarrow second\_letter = "L"$ 
|   |—  $35 \leq LL < 50 \rightarrow second\_letter = "I"$ 
|   |—  $50 \leq LL < 70 \rightarrow second\_letter = "H"$ 
|   |—  $70 \leq LL < 90 \rightarrow second\_letter = "V"$ 
|   |—  $LL \geq 90 \rightarrow second\_letter = "E"$ 
|
|— RETURN  $first\_letter + second\_letter$ 
```

Edge Cases — Handle These Before Classification

Think of these as input guards that run before the main logic:

Condition	What It Means	What to Return
$PL \geq LL$	Invalid inputs	Throw a validation error
$PI < 4$	Barely plastic soil	Flag as borderline, classify with caution
$PI = 0$ or non-plastic	No plastic behaviour	Return "NP" — skip classification
LL exactly on a boundary	Ambiguous range	Define a rule — e.g. $LL = 50$ goes to "H" not "I"
$PI > 0.9 \times (LL - 8)$	Likely a test error (above U-line)	Flag as suspect, do not classify

What to Return to the UI

Think of the output as a structured object with everything the front end needs:

```
{
  inputs: { LL, PL },
  derived: { PI, A_line_value },
```

```
position: "Above A-line" or "Below A-line",
BS_classification: "CI",
USCS_classification: "CL",
plasticity_description: "Clay of Intermediate Plasticity",
engineering_note: "Moderate volume change potential"
}
```

Summary

The classification is nothing more than two comparisons and a string join:

- Compare PI against the A-line value to get the soil type (C or M)
- Look up the LL range to get the plasticity descriptor (L, I, H, V, or E)
- Concatenate the two letters to get the final BS classification code

For USCS, skip the second step entirely — just use four fixed conditions combining the A-line check and the $LL < 50$ boundary.